

SHIZUHAMA AB, Japan

WMO: 476580

Lat: 34.813N

Lon: 138.298E

Elev: 33

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.6	(c) 33.4	(d) 10.2	(e) 9.3	(f) 40.5	(g) 13.8	(h) 11.1	(i) 40.0	(j) 27.0	(k) 47.1	(l) 25.0	(m) 46.3	(n) 9.3	(o) 270	(p) 0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.9	(c) 90.0	(d) 78.7	(e) 88.0	(f) 78.6	(g) 86.1	(h) 77.8	(i) 80.7	(j) 86.4	(k) 79.9	(l) 85.3	(m) 79.1	(n) 84.1	(o) 13.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.1	151.2	83.9	78.6	148.7	83.5	77.3	142.0	82.3	44.5	86.9	43.5	85.7	42.7	84.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.7	(b) 21.4	(c) 19.2	(e) 27.2	(f) 95.1	(g) 1.7	(h) 2.7	(i) 26.0	(j) 97.0	(k) 25.0	(l) 98.6	(m) 24.1	(n) 100.1	(o) 22.8	(p) 102.0		
(4) 23.7	(5) 21.4	(6) 19.2	(7) 23.1	(8) 82.5	(9) 1.5	(10) 1.3	(11) 22.0	(12) 83.5	(13) 21.1	(14) 84.2	(15) 20.2	(16) 85.0	(17) 19.1	(18) 85.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	43.7	45.1	50.2	58.4	66.0	71.9	78.8	81.1	75.7	66.9	57.3	48.1	(6)
(7)		DBStd	13.54	4.32	4.90	5.13	5.31	3.88	3.93	3.99	3.24	4.32	4.53	5.12	5.12	(7)
(8)		HDD50	517	200	151	61	4	0	0	0	0	0	0	4	97	(8)
(9)		HDD65	2702	659	557	458	206	33	1	0	0	0	30	235	523	(9)
(10)		CDD50	4909	6	15	69	255	497	656	892	963	770	524	223	39	(10)
(11)		CDD65	1617	0	0	0	7	65	207	427	498	320	89	4	0	(11)
(12)		CDH74	12019	0	0	0	25	198	774	3567	4990	2179	279	7	0	(12)
(13)		CDH80	3264	0	0	0	1	21	133	1010	1608	461	30	0	0	(13)
(14)	Wind	WSAvg	8.2	10.3	9.5	9.5	8.7	7.6	7.2	7.4	7.1	6.8	7.0	7.7	9.7	(14)
(15)	Precipitation	PrecAvg	85.7	3.2	4.3	7.1	7.8	8.4	9.9	9.7	7.0	10.7	8.8	5.7	3.1	(15)
(16)		PrecMax	116.1	7.4	9.2	12.3	14.9	12.2	18.1	22.2	26.7	23.1	35.4	12.9	7.2	(16)
(17)		PrecMin	59.4	0.0	1.1	2.6	1.9	4.2	3.6	1.0	0.6	3.2	0.9	0.5	0.0	(17)
(18)		PrecStd	16.3	2.0	2.3	2.4	3.1	2.6	3.2	5.4	4.9	5.0	6.5	3.0	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.1	65.8	69.7	77.1	82.7	88.0	93.1	94.9	89.9	83.8	74.9	66.4	(19)
(20)			MCWB	51.5	54.8	56.5	62.2	66.8	75.5	78.4	79.5	77.5	71.2	64.3	59.4	(20)
(21)		2%	DB	57.2	60.6	64.8	73.1	78.7	83.9	89.7	91.2	86.4	79.2	71.2	62.4	(21)
(22)			MCWB	48.7	52.2	54.0	60.9	65.7	73.8	78.7	79.3	76.8	69.2	61.8	54.0	(22)
(23)		5%	DB	55.0	57.1	62.5	69.9	75.5	80.6	87.6	89.2	84.4	77.0	68.2	59.3	(23)
(24)			MCWB	46.5	48.2	54.0	60.5	65.5	72.3	78.5	79.1	75.8	68.3	59.9	50.9	(24)
(25)		10%	DB	52.0	53.9	60.1	67.7	73.5	78.4	85.7	86.4	82.5	75.0	66.2	57.2	(25)
(26)			MCWB	43.3	45.9	52.3	59.3	65.0	71.7	77.8	78.5	75.2	67.0	58.5	49.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	59.6	61.4	69.5	73.9	78.1	81.7	82.2	80.8	75.3	68.4	61.9	(27)
(28)			MCDWB	57.5	62.5	64.0	72.0	76.9	83.6	86.9	88.0	85.1	79.2	71.3	64.8	(28)
(29)		2%	WB	50.5	54.0	58.9	65.1	70.1	76.3	80.3	81.0	79.2	72.4	64.2	56.1	(29)
(30)			MCDWB	54.7	59.1	62.3	69.2	73.9	81.0	86.1	86.8	83.4	77.1	68.0	60.1	(30)
(31)		5%	WB	47.2	50.5	56.1	62.9	68.2	74.7	79.5	80.1	78.1	70.2	62.0	52.9	(31)
(32)			MCDWB	52.2	54.4	60.6	67.3	72.8	78.7	84.9	85.7	82.3	74.9	66.4	57.5	(32)
(33)		10%	WB	44.8	47.5	53.3	60.8	66.6	72.9	78.6	79.5	76.8	68.2	59.9	50.5	(33)
(34)			MCDWB	50.6	52.1	58.4	65.1	71.5	76.9	83.6	84.8	80.9	73.0	64.6	55.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.2	15.6	15.6	14.6	13.3	10.4	10.3	10.9	10.9	12.5	14.2	15.0	(35)
(36)			MCDBR	17.4	18.6	19.3	18.1	17.0	14.2	13.9	13.8	12.9	14.6	16.3	16.6	(36)
(37)			MCWBR	11.8	13.1	13.2	11.3	9.6	7.0	6.2	6.1	6.9	8.5	10.7	11.3	(37)
(38)		5% WB	MCDBR	15.3	15.4	16.0	14.3	13.7	11.7	11.9	11.6	11.2	12.2	13.5	14.0	(38)
(39)			MCWBR	12.3	13.0	13.8	10.5	9.4	6.7	6.1	6.0	7.2	8.6	10.9	11.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.373	0.423	0.483	0.533	0.547	0.567	0.556	0.544	0.497	0.472	0.413	0.373	(40)	
(41)		taud	2.269	2.130	1.979	1.904	1.936	1.958	2.037	2.059	2.162	2.150	2.234	2.292	(41)	
(42)		Ebn at Noon	257	255	249	244	242	236	238	238	243	237	240	249	(42)	
(43)		Edh at Noon	34	43	54	61	60	59	54	52	45	42	35	31	(43)	
(44)	All-Sky Solar Radiation	RadAvg	875	1065	1341	1537	1660	1462	1625	1664	1308	1028	843	777	(44)	
(45)		RadStd	58	88	86	163	171	142	223	134	114	106	73	56	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page